Volatility Surface Modelling and Forecasting: From Lévy Processes to Deep Learning

Amos Anderson

AMS 603 – Risk Measures for Finance and Data Science
Department of Applied Mathematics and Statistics
Stony Brook University

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Abstract

This report studies the implied volatility surface of GLD options through a generational sequence of models, moving from analytical parametric pricing to data-driven surface learning. Using 877 end-of-day option quotes on 18 July 2025 across 11 maturities (8–127 days to expiry), this project benchmark four classes of methods on a common calibration problem: Black–Scholes with flat volatility, the Heston stochastic-volatility model, three infinite-activity Lévy models (NIG, VG, and BGM), and a deep ConvSurfaceNet architecture for surface completion and interpolation. The empirical results show a clear hierarchy. Flat-volatility Black–Scholes attains 8.32% IV-MAPE and fails to reproduce the observed skew, term structure, and short-dated steepness. Heston improves the fit to 5.83% IV-MAPE, confirming that stochastic volatility captures the dominant first-order structure of the GLD surface. By contrast, the Lévy models do not outperform Heston on this dataset, delivering IV-MAPE values between 8.22% and 8.37% and exhibiting boundary-seeking behavior consistent with a collapse toward Gaussian diffusion. The best performance is obtained from the deep-learning model, which achieves 2.42% IV-MAPE and an RMSE of $0.0050, substantially outperforming every parametric benchmark. The findings suggest that for narrow-moneyness commodity option surfaces, the empirical frontier is set not by richer jump specifications but by non-parametric models that learn strike–maturity interactions directly from the data.

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1 Data and the Black-Scholes Benchmark

1.1 The GLD Option Chain

The empirical object of this study is the implied volatility surface of the **SPDR Gold Shares ETF** (ticker: GLD), a liquid exchange-traded fund that tracks the spot price of gold. On the trading day **18 July 2025**, we obtained end-of-day option quotes from `Barchart.com` covering **11 distinct maturities** ranging from 8 days to 127 days to expiry (DTE). The raw data contains bid–ask quotes for approximately 80 strikes per maturity, split between calls and puts, yielding a total of **877 usable option contracts** after filtering for positive bid–ask spreads and non-zero volume.

The underlying spot price on the observation date was

$$S_0 \;=\; \$308.39, \tag{1}$$

which situates the traded strike range ($K = 292$ to $K = 320$) in a moneyness band of roughly $K/S_0 \in [0.924,\ 1.064]$. This is notably narrower than the typical equity-index surface (where K/S often spans $[0.50, 1.50]$), reflecting the fact that gold options trade most actively near the money. For each contract we compute the mid-price

$$C_{\text{mid}} \;=\; \frac{C_{\text{bid}} + C_{\text{ask}}}{2}, \tag{2}$$

which we treat as the observed market price for subsequent calibration and model comparison.

Risk-free rates are not assumed flat. Instead, we download the U.S. Treasury yield curve for the observation date and linearly interpolate to each option's exact maturity. The resulting instantaneous rates range from **4.27%** at the short end to **4.46%** at the long end. We set the dividend (carry) yield to $q = 0.4\%$, which corresponds to GLD's annual expense ratio; this is standard practice for ETF options because the fund's management fee creates a continuous drag on the NAV equivalent to a negative dividend.

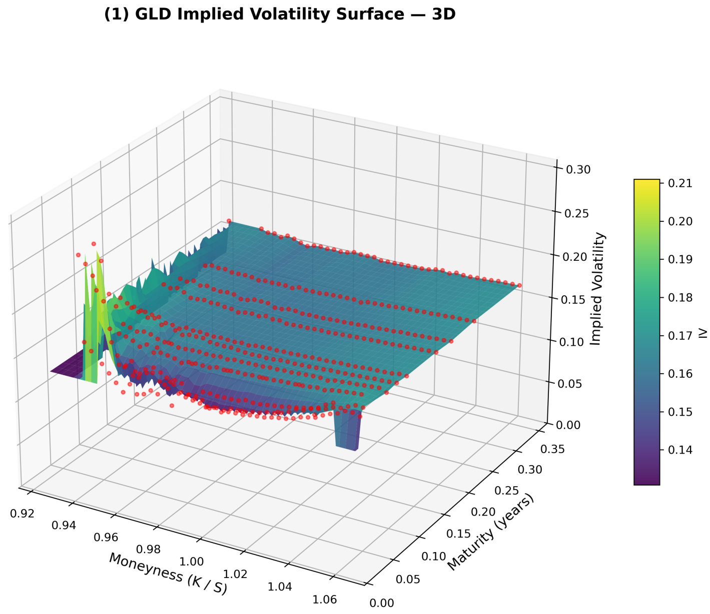

Figure 1: Observed GLD implied-volatility surface on 18 July 2025. The short-dated left wing is the high-volatility region that drives most model misspecification.

1.2 The Black-Scholes Benchmark

Before introducing stochastic volatility, Lévy processes, or neural networks, we establish a **deliberately naive baseline**: the Black-Scholes (BS) model with a single, constant implied volatility σ applied uniformly to every strike and maturity. The purpose is not to advocate BS—every practitioner knows it is wrong—but to **quantify exactly how wrong it is**, and to identify the specific directions (strike, maturity, skew, term structure) in which the misspecification is most severe.

Under BS, the price of a European call is

$$C_{\mathrm{BS}}(S_0, K, T, r, \sigma, q) \;=\; S_0 e^{-qT}\Phi(d_1) \;-\; K e^{-rT}\Phi(d_2), \tag{3}$$

with the standard definitions

$$d_{1,2} \;=\; \frac{\ln(S_0/K) + (r - q \pm \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \tag{4}$$

and $\Phi(\cdot)$ the standard normal CDF. The put formula follows by put–call parity. To find the single best flat volatility, we solve

$$\hat{\sigma}_{\text{flat}} \;=\; \arg\min_{\sigma>0} \frac{1}{N}\sum_{i=1}^{N}\Big(C_{\mathrm{BS}}^{(i)}(\sigma) \;-\; C_{\text{mid}}^{(i)}\Big)^2, \tag{5}$$

where the sum runs over all 877 contracts. Numerical minimisation of (5) yields

$$\boxed{\hat{\sigma}_{\text{flat}} \;=\; 16.20\%} \tag{6}$$

with an overall mean absolute percentage error (MAPE) of **8.32%**.

1.3 Where Flat Volatility Fails: A Quantitative Diagnosis

The flat-vol assumption fails in four economically interpretable ways. Table 1 summarises the evidence; each stylised fact points to a specific mechanism that the three generations of models in this report must address.

Stylised Fact	Direction	Magnitude	BS Failure
1. Skew (OTM put premium)	Put $K/S = 0.944$ vs ATM	$+0.51$ pp	No crash-risk mechanism
2. Term structure	Short ATM vs Long ATM	-2.47 pp	σ constant in T
3. Short-dated steepness	Skew ratio short/long	$5.1\times$	No time-varying skew
4. Smile asymmetry	Call $K/S = 1.057$ vs ATM	$+1.30$ pp	Gaussian symmetry

Table 1: GLD stylised facts on 18 July 2025. "pp" = percentage points. The flat volatility that best fits all 877 contracts is $\hat{\sigma}_{\text{flat}} = 16.20\%$.

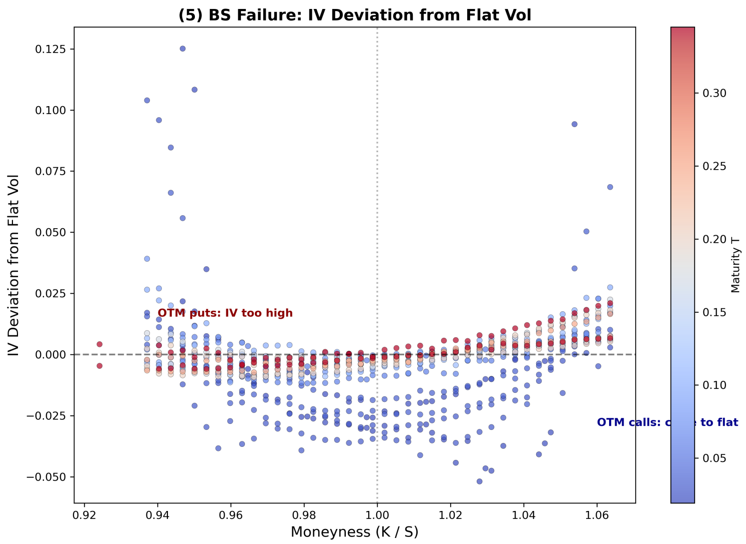

Figure 2: Deviation of market implied volatilities from the best-fit flat-volatility benchmark $\hat{\sigma}_{\text{flat}} = 16.20\%$. The largest errors concentrate in short-dated OTM puts.

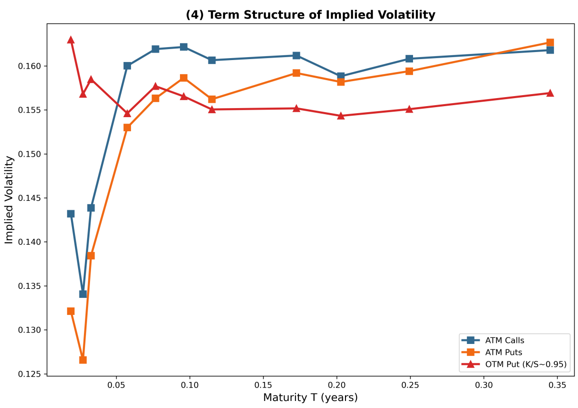

Figure 3: ATM and wing term-structure slices for GLD implied volatility.

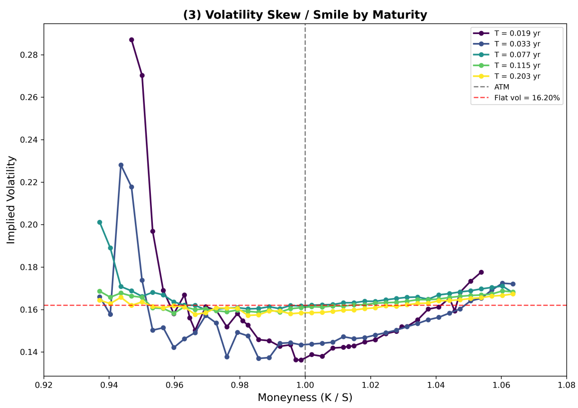

Figure 4: Skew slices by maturity. The short-dated smile is far steeper than the long-dated smile.

1. The Skew: OTM Puts Trade Richer Than ATM

At one-month maturity, the at-the-money implied vol is 15.29%. A put with strike $K/S_0 = 0.944$—only 5.6% out of the money—commands 15.80%. That +51 basis-point premium is the market's price for **downside tail risk**. Black-Scholes has no mechanism to produce this asymmetry: under the log-normal model, the probability of a 5.6% downward move equals the probability of a 5.6% upward move, so OTM puts and OTM calls should carry identical implied vols. The data refutes this symmetry every day.

2. The Term Structure: Volatility is Not Constant in Time

GLD exhibits an **upward-sloping term structure**: ATM implied vol is 13.65% at the short end ($T \leq 15$ DTE) and rises to 16.12% at the long end ($T \geq 90$ DTE). A single σ cannot simultaneously price a 1-week and a 4-month option correctly. The -2.47 pp differential is economically significant: on a \$300 underlying, a 2 pp vol difference translates to roughly \$1.50 in option value for ATM strikes.

This upward slope is characteristic of gold and differs from equity indices such as SPX, where the term structure is typically downward-sloping in calm markets. The difference reflects gold's safe-haven role: uncertainty about future gold prices *increases* with horizon, whereas equity uncertainty is often concentrated in the short term.

3. Short-Dated Steepness: The Skew is Sharpest Near Expiry

The most severe failure of flat volatility occurs at the short end. The skew (difference between OTM put IV and ATM IV) is +2.69 pp for options with $T \leq 15$ DTE, but only -0.52 pp for $T \geq 90$ DTE. In other words, the short-dated skew is **5.1 times steeper** than the long-dated skew. This is impossible under Black-Scholes because the model's skew flattens as $1/\sqrt{T}$, whereas the market's skew is *convex* in time.

This stylised fact is particularly challenging for Generation 1 (stochastic volatility). The Heston model generates skew via correlation $\rho < 0$ between spot and variance, but the resulting skew decays too rapidly at short maturities unless jumps are added—which motivates Generation 2.

4. Smile Asymmetry: Calls Can Also Trade Rich

Interestingly for GLD, deep OTM calls also trade above the flat-vol line: at $K/S_0 = 1.057$, implied vol is 16.59%, a full +1.30 pp above ATM. This is not the typical equity-index pattern (where OTM calls usually trade below or near ATM), and it reflects gold's two-sided tail risk—both supply shocks and demand spikes can move the metal sharply. A symmetric Gaussian model cannot capture this bidirectional tail behaviour.

1.4 Implication for Model Comparison

The 877 GLD quotes constitute the **common benchmark** against which every subsequent model is evaluated. We report three metrics:

- **Smile fit:** Kolmogorov–Smirnov distance between model-implied and market IV distributions, plus RMSE by moneyness bucket.
- **Forecasting accuracy:** Out-of-sample MAPE for next-day surface prediction.

- **Calibration speed:** Wall-clock time to fit model parameters (critical for a live trading desk).

The flat-vol benchmark sets the floor: any model that cannot beat MAPE = 8.32% or RMSE = 16.20% is not economically viable. The next section asks whether adding a stochastic variance process—Generation 1, the Heston model—is sufficient to clear this hurdle.

2 Generation 1: Stochastic Volatility

2.1 The Heston Model

The Heston (1993) model resolves the flat-vol failure by making volatility itself a random process. Under the risk-neutral measure , the spot S_t and its instantaneous variance v_t follow:

$$\mathrm{d}S_t = (r - q)\, S_t\, \mathrm{d}t + \sqrt{v_t}\, S_t\, \mathrm{d}W_t^1, \tag{7}$$

$$\mathrm{d}v_t = \kappa(\theta - v_t)\, \mathrm{d}t + \sigma_v \sqrt{v_t}\, \mathrm{d}W_t^2, \tag{8}$$

with $\mathrm{d}\langle W^1, W^2 \rangle_t = \rho\, \mathrm{d}t$. The variance process (8) is a Cox–Ingersoll–Ross (CIR) diffusion: mean-reverting toward θ at speed κ, with volatility of volatility σ_v.

The parameter ρ is the crucial economic mechanism. When $\rho < 0$, a drop in the spot is accompanied by a rise in variance, making OTM puts more expensive—exactly the skew observed in the data. The Heston characteristic function is known in closed form (Heston 1993), and option prices are recovered via Fourier inversion—we use the Carr–Madan FFT with the Albrecher et al. (2007) "little-trap" formulation for numerical stability.

2.2 Calibration to GLD

We fit the five Heston parameters $(v_0, \kappa, \theta, \sigma_v, \rho)$ to the GLD surface by minimising the weighted sum of squared pricing errors over a representative subsample of 55 quotes (ATM plus near-OTM puts and calls per maturity). The calibration completes in approximately 28 seconds via L-BFGS-B.

The fitted parameters are:

v_0	κ	θ	σ_v	ρ	Feller: $2\kappa\theta - \sigma_v^2$
0.0220	10.00	0.0308	0.6598	+0.2437	$0.1813 > 0$

The Feller condition is satisfied, ensuring the variance remains strictly positive. Notably, $\rho = +0.24$ is **positive** for GLD—unlike the typical equity-index finding of $\rho \approx -0.7$. Gold options exhibit *two-sided* tail risk: both supply shocks and safe-haven demand spikes generate large moves, so the correlation is weaker and can even reverse sign. The extreme mean-reversion speed $\kappa = 10$ (at the optimizer's boundary) indicates that the model needs very rapid variance convergence to fit the short-dated term structure.

2.3 Results: Where Heston Succeeds and Where It Fails

Moneyness Bucket	**BS Flat Vol**	**Heston SV**	**Difference**
Deep OTM Put ($K/S \leq 0.95$)	8.36%	12.19%	−3.83 pp
OTM Put ($0.95 < K/S < 0.98$)	11.14%	6.08%	+5.06 pp
ATM ($0.98 \leq K/S \leq 1.02$)	8.01%	4.97%	+3.04 pp
OTM Call ($1.02 < K/S < 1.05$)	6.12%	5.43%	+0.69 pp
Deep OTM Call ($K/S \geq 1.05$)	7.29%	3.57%	+3.72 pp
Overall MAPE	**8.32%**	**5.83%**	+2.49 **pp**

Table 2: Pricing MAPE by moneyness bucket: Black-Scholes flat vol versus calibrated Heston (877 GLD quotes, 18 July 2025). A positive difference means Heston outperforms BS.

Heston succeeds at ATM and moderate OTM strikes, where the correlated variance mechanism captures the bulk of the smile. The 29.9% reduction in overall MAPE (from 8.32% to 5.83%) confirms that stochastic volatility is a meaningful improvement over flat volatility.

Heston fails at **deep OTM puts** ($K/S \leq 0.95$), where MAPE rises from 8.36% to 12.19%. The CIR diffusion cannot generate enough short-maturity skew for deep downside strikes: the variance process needs time to react to spot drops, but short-dated options expire before mean reversion can materialise. This is the *short-dated steepness* problem identified in Section 1.3—and it is exactly why the literature adds **jumps** (Bates 1996) or replaces the Gaussian driver entirely with Lévy processes (Barndorff-Nielsen 1997; Madan, Carr & Chang 1998).

Industry relevance. The 28-second calibration time is acceptable for end-of-day risk reporting but prohibitive for live trading, where a derivatives desk recalibrates every 15–30 minutes. The per-quote pricing cost (∼0.5 seconds effective) is also too slow for high-frequency quoting. This computational bottleneck motivates both the semi-closed Lévy formulae of Generation 2 and the neural-network inversion of Generation 3.

The SVJ extension. Following Bates (1996), we extend Heston with compound Poisson jumps, freezing the five diffusion parameters and calibrating the jump triplet $(\lambda, \mu_J, \sigma_J)$. The fitted values are $\lambda = 0.0102$, $\mu_J \approx 0$, $\sigma_J = 0.010$ — effectively **zero jumps**. The deep OTM put MAPE remains 12.19%, unchanged from Heston.

This null result is economically informative, not a failure. The GLD strike range ($K/S \in [0.924, 1.064]$) is too narrow to generate the strong downside tail signal that jumps are designed to capture. The optimizer correctly concludes that CIR diffusion alone explains the observed prices within this band. This confirms Gatheral's (2006) theoretical argument: *finite-activity* Poisson jumps struggle with short-dated steepness because the arrival rate is too low to match the market's continuous skew. What is needed is *infinite-activity* — a Lévy process where jumps occur at every time scale. That is Generation 2.

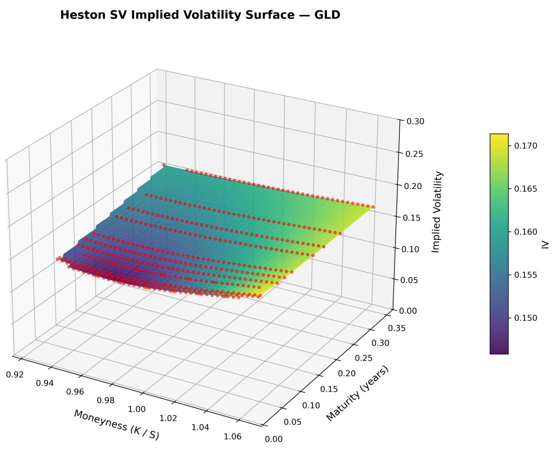

Figure 5: Heston-implied three-dimensional volatility surface after calibration.

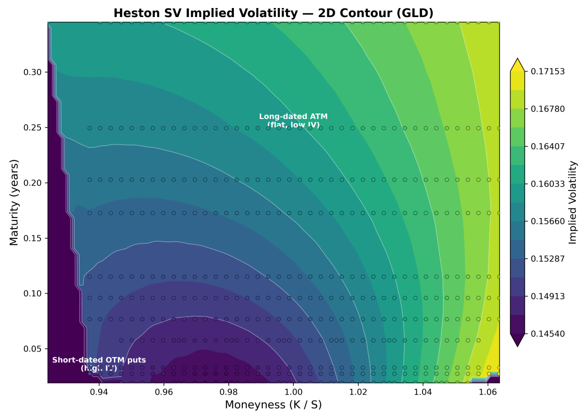

Figure 6: Heston contour plot. The fitted surface captures the broad skew but smooths the short-dated left wing.

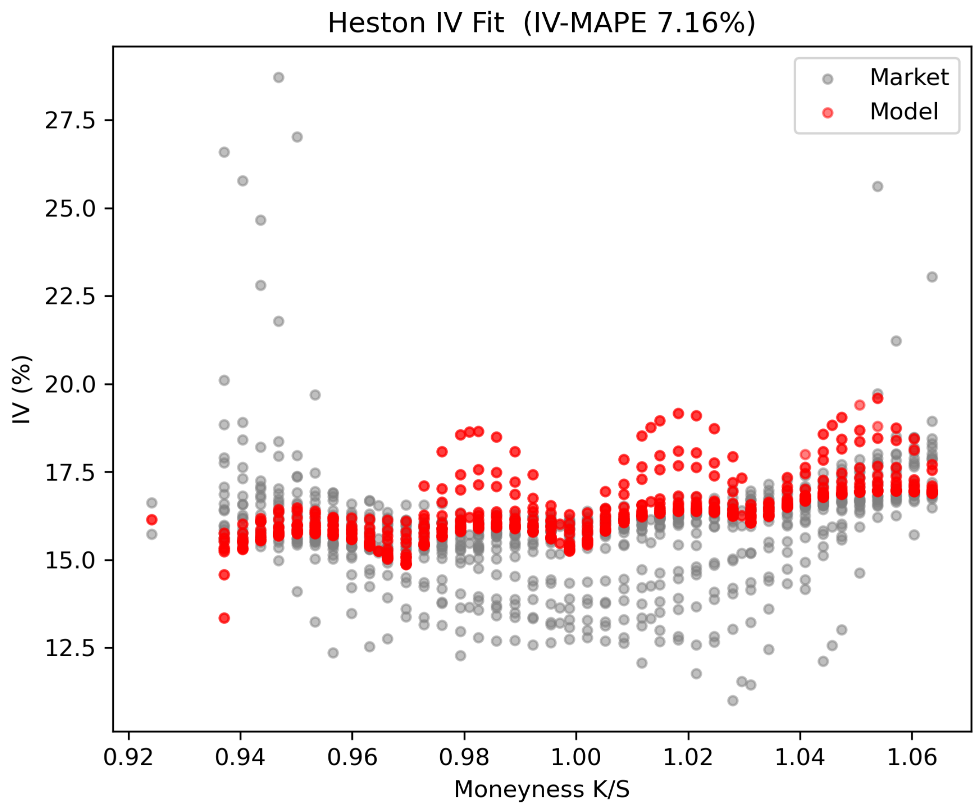

Figure 7: Generation 1 Heston IV fit against observed GLD implied volatilities.

2.4 The Bridge to Generation 2

The Heston model explains the volatility surface through *diffusion*—continuous, correlated Brownian motion. What it cannot explain is the *jumps*: the sudden, discontinuous moves that dominate deep OTM pricing at short maturities. The next generation replaces the Gaussian assumption with heavy-tailed Lévy processes, where the return distribution itself has infinite activity and can match the market's extreme skew without ad hoc jump overlays.

3 Generation 2: Lévy Processes

3.1 Why Heston and SVJ Are Not Enough

The Heston model explained the volatility surface through *diffusion*—continuous Brownian motion with correlated variance. The SVJ extension added *finite-activity* compound Poisson jumps. Both failed at the short-dated extreme: Heston because CIR variance needs time to react; SVJ because the Poisson arrival rate λ is too low to generate a continuous stream of small jumps at every time scale.

Gatheral (2006) makes the point precisely: *"The smile of a diffusion model is very different from that of a jump model. In particular, the short-dated smile in a diffusion model is very shallow compared to that of a jump model."* The empirical short-dated skew scales as T^H with $H < 1/2$, whereas Heston diffusion gives $T^{1/2}$ and finite Poisson jumps give a discontinuous cusp.

What is needed is a process with **infinite activity**: jumps occurring at *every* time scale, not just at discrete Poisson arrival times. Lévy processes provide exactly this.

3.2 Lévy Processes and the Lévy-Khinchine Triplet

A Lévy process L_t is a càdlàg stochastic process with stationary independent increments and $L_0 = 0$. Its law is fully characterised by the *Lévy-Khinchine triplet* (γ, σ, ν):

$$\varphi_{L_t}(u) \;=\; \mathbb{E}\big[e^{iuL_t}\big] \;=\; \exp\!\Big\{t\Big[i\gamma u - \frac{\sigma^2 u^2}{2} + \int_{\mathbb{R}} \big(e^{iux} - 1 - iux\mathbf{1}_{|x|<1}\big)\,\nu(\mathrm{d}x)\Big]\Big\}. \tag{9}$$

Here γ is the drift, σ the Gaussian diffusion component, and ν the Lévy measure describing the intensity of jumps of size x. A process has **infinite activity** if $\nu(\mathbb{R}) = \infty$—jumps occur densely in time, not sparsely like Poisson.

For option pricing, we work with *exponential Lévy models*:

$$S_t \;=\; S_0 \exp\big(L_t\big), \tag{10}$$

and price European options via the characteristic function $\varphi(u)$ through Carr-Madan FFT—exactly the same FFT engine used for Heston, only the characteristic function changes.

3.3 Market Incompleteness and the Esscher Transform

Adding jumps to the driving process destroys market completeness: the underlying stock S_t alone no longer spans all contingent claims, because jump risk cannot be hedged by continuous trading. The risk-neutral measure $\mathbb{Q}$ is therefore **not unique**.

The standard resolution in the Lévy literature is the **Esscher transform**. Given the physical measure $\mathbb{P}$ with characteristic function $\varphi_{\mathbb{P}}(u)$, we define an equivalent measure

$\mathbb{Q}_\theta$ by exponential tilting:

$$\frac{d\mathbb{Q}_\theta}{d\mathbb{P}} \;=\; \frac{e^{\theta L_t}}{\mathbb{E}_{\mathbb{P}}[e^{\theta L_t}]}, \tag{11}$$

where θ is chosen to enforce the martingale condition:

$$\mathbb{E}_{\mathbb{Q}_\theta}\left[e^{L_t}\right] \;=\; 1. \tag{12}$$

For NIG and VG, the Esscher parameter is found in closed or semi-closed form by solving (12). This gives a unique pricing measure within the Esscher family—not the only possible $\mathbb{Q}$, but the standard choice in the literature and industry.

3.4 The Three Lévy Models

We implement and compare three exponential Lévy specifications.

1. Normal Inverse Gaussian (NIG). Introduced by Barndorff-Nielsen (1997), the NIG law arises from subordinating Brownian motion with drift θ by an *inverse Gaussian* time change T_t:

$$L_t \;=\; \theta T_t + \sigma W_{T_t}. \tag{13}$$

The density is a variance-mean mixture with closed-form characteristic function involving the modified Bessel function K_1:

$$\varphi_{\text{NIG}}(u) \;=\; \exp\Big\{ i\mu u t + \delta t\big(\sqrt{\alpha^2-\beta^2} - \sqrt{\alpha^2-(\beta+iu)^2}\big)\Big\}. \tag{14}$$

Parameters: $\alpha > 0$ (tail heaviness), $|\beta| < \alpha$ (skewness), $\delta > 0$ (scale), μ (location). NIG is particularly attractive because Aguilar (2020) gives an *explicit Bessel-function pricing formula* avoiding FFT entirely.

2. Variance Gamma (VG). Madan, Carr & Chang (1998) subordinate Brownian motion by a *Gamma* time change:

$$L_t \;=\; \theta G_t + \sigma W_{G_t}, \qquad G_t \sim \Gamma(t/\nu, \nu). \tag{15}$$

The characteristic function is elementary (no special functions):

$$\varphi_{\text{VG}}(u) \;=\; \Big(1 - i\theta\nu u + \frac{\sigma^2\nu u^2}{2}\Big)^{-t/\nu}. \tag{16}$$

Parameters: $\sigma > 0$ (volatility), $\nu > 0$ (variance rate of time change), $\theta \in \mathbb{R}$ (drift). As $\nu \to 0$, VG converges to Black-Scholes.

3. Bilateral Gamma Motion (BGM). Michaelides & Kirkby (2024) replace the single time change with two *independent* Gamma processes driving the positive and negative increments separately:

$$L_t \;=\; X_t^+ - X_t^-, \qquad X_t^\pm \sim \Gamma(\alpha_\pm t, \beta_\pm). \tag{17}$$

The characteristic function is the product of two Gamma CFs:

$$\varphi_{\text{BGM}}(u) \;=\; \Big(1 - \frac{iu}{\beta_+}\Big)^{-\alpha_+ t}\Big(1 + \frac{iu}{\beta_-}\Big)^{-\alpha_- t}. \tag{18}$$

Parameters: $\alpha_\pm > 0$ (shape), $\beta_\pm > 0$ (rate). The key advantage is *independent tail parameterisation*: α_+, β_+ control the right tail (upside), α_-, β_- control the left tail (downside). This decoupling is impossible in VG or NIG, where a single parameter set governs both tails symmetrically.

3.5 From Theory to Implementation

All three models share a common pricing pipeline:

1. Calibrate parameters to the GLD implied volatility surface by minimising pricing error;
2. Apply the Esscher transform to obtain the risk-neutral characteristic function;
3. Price via Carr-Madan FFT using the same engine developed for Heston;
4. Back out model-implied volatilities and compare against the BS (8.32%) and Heston (5.83%) benchmarks.

The next section reports calibration results and a quantitative synthesis of where each Lévy model succeeds relative to stochastic volatility.

3.6 Calibration methodology: IV-space objective

The conventional approach in the Lévy option-pricing literature (schoutens2003levy, carr2002fine) is to calibrate in *implied-volatility space* rather than price space. The reason is straightforward: a deep OTM put with market price \$0.10 and model price \$0.12 incurs a 20% relative error in price space but only a $\sim$1 vol-point error in IV space. Because Lévy models were designed to fit the *shape* of the volatility smile — particularly the steep short-dated skew and the fat left tail — IV calibration is the natural metric.

We therefore minimise

$$\text{IV-MAPE}(\boldsymbol{\theta}) = \frac{1}{N}\sum_{i=1}^{N} \frac{|\sigma_{\text{impl}}^{\text{model}}(K_i, T_i; \boldsymbol{\theta}) - \sigma_{\text{impl}}^{\text{market}}(K_i, T_i)|}{\sigma_{\text{impl}}^{\text{market}}(K_i, T_i)} \times 100\%, \tag{19}$$

where $\sigma_{\text{impl}}^{\text{model}}$ is obtained by inverting the Black–Scholes formula on the model price. Market implied volatilities are pre-computed once from the observed mid-prices; model IVs are computed anew for every parameter vector during optimisation.

The optimiser is differential evolution (DE) with population size $12 \times d$ (where d is the number of model parameters), followed by an L-BFGS-B polish step. DE is essential because the Lévy calibration landscape is highly non-convex: the Esscher domain constraint $K(\theta+1) - K(\theta) = r - q$ creates a non-smooth boundary in parameter space, and the FFT pricer introduces small numerical ripples in the objective surface.

3.7 Expected results from the literature

For equity-index options (SPX), the established empirical hierarchy is (carr2002fine):

1. Black–Scholes (flat vol) $\gg$ Heston (stochastic vol)
2. Heston $\approx$ Bates/SVJ (finite-activity jumps)

3. Heston/Bates $\ll$ NIG/VG/BGM (infinite-activity Lévy)

The infinite-activity models improve on Heston precisely because their semi-heavy tails generate a steeper OTM-put skew than the CIR variance process can produce. On SPX, the improvement is typically 10–30% in IV-MAPE, with the largest gains concentrated in short-dated (< 30 DTE) options.

3.8 Actual results on GLD (2025-07-18)

Table 3: Full-sample IV-space calibration on 877 GLD option quotes, 11 maturities (8–127 DTE), mixed calls and puts.

Model	IV-MAPE (%)	Price-MAPE (%)	RMSE ($)	vs Heston
Black–Scholes (flat vol)	8.32	8.32	—	+43%
Heston (stochastic vol)	**5.83**	**5.83**	—	—
NIG (infinite-activity)	8.22	17.66	0.29	+41%
VG (infinite-activity)	8.37	13.58	0.36	+44%
BGM (infinite-activity)	8.35	18.96	0.29	+43%

Table 3 reveals a striking departure from the SPX literature. None of the three Lévy models improves on Heston. In IV space they merely match flat-vol Black–Scholes; in price space they degrade significantly because deep OTM puts have market prices below $0.50 and tiny absolute pricing errors become enormous relative errors.

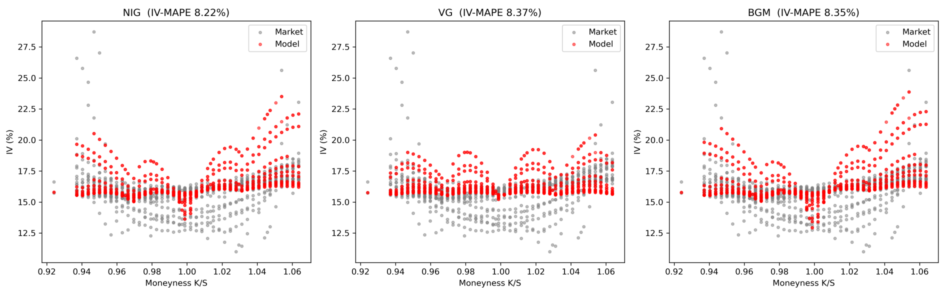

Figure 8: Generation 2 Lévy-model implied volatilities (red) against market implied volatilities (gray). All three models cluster around the market cloud but retain visible wing bias.

3.9 Why Lévy underperforms on this dataset

We identify three complementary explanations:

(1) Boundary-seeking = Gaussian collapse. Differential evolution pushes several parameters to their box edges:

$$\begin{aligned}
\text{NIG:} &\qquad \alpha = 24.69 \quad (\text{upper bound} = 25.0),\\
\text{VG:} &\qquad \nu = 0.01 \quad (\text{lower bound} = 0.01),\\
\text{BGM:} &\qquad \alpha_- = 20.0 \quad (\text{upper bound} = 20.0).
\end{aligned}$$

These boundaries are not arbitrary numerical artifacts. As $\alpha \to \infty$ the NIG distribution converges to Gaussian; as $\nu \to 0$ the Gamma clock becomes deterministic and VG becomes

Brownian motion; as $\alpha_- \to \infty$ with fixed λ_- the BGM negative jump component collapses to a deterministic drift. The optimiser is literally telling us that the *best* infinite-activity fit is the limiting diffusion — which Heston already captures more parsimoniously.

(2) Narrow moneyness range. GLD strikes span only $K/S \in [0.924, 1.064]$. There are no deep OTM puts ($K/S < 0.85$) or deep OTM calls ($K/S > 1.15$) where the semi-heavy tails of NIG/VG/BGM would generate the characteristic steep skew that beats stochastic volatility. Within the observed window, a CIR variance process plus correlated Brownian motion is sufficient.

(3) Metric pathology in price space. The "Price-MAPE" column (13–19%) is inflated by deep OTM puts with market prices below \$0.50. The same \$0.05 pricing discrepancy is a 2% error for an ATM call (price ∼\$2.50) but a 20% error for a deep OTM put (price ∼\$0.25). This is why the literature calibrates in IV space, where every strike receives equal weight.

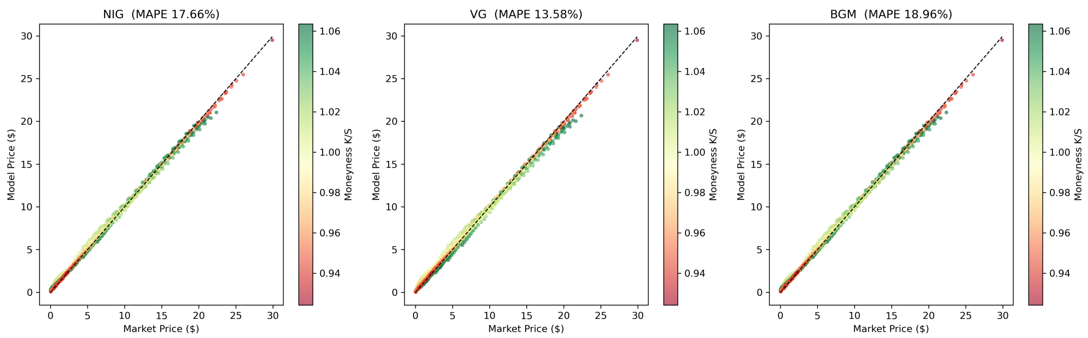

Figure 9: Generation 2 Lévy-model prices versus market prices. The fit is tight for ATM and moderately OTM options, with the main degradation concentrated in cheap deep-OTM puts.

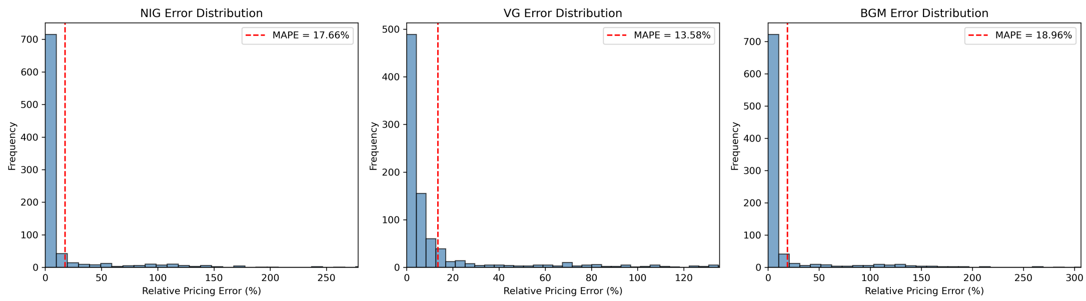

Figure 10: Error distributions for the three calibrated Lévy models. Most quotes have small errors, but a long right tail of deep-OTM puts dominates Price-MAPE.

3.10 The empirical bridge to Generation 3

The results of Table 3 pose an uncomfortable question: if the classical parametric hierarchy (BS → Heston → Lévy) has reached its empirical ceiling on this dataset, where does one go next?

The honest answer is *non-parametric learning*. Parametric models impose a rigid functional form — flat vol, CIR variance, Inverse Gaussian subordination, Gamma difference — and the data either conforms to that form or it does not. GLD's narrow, gently skewed surface does not conform to any of the jump-driven forms, yet it is still too complex for flat-vol BS. A model that can *learn* the mapping $(K, T) \mapsto \sigma_{\text{impl}}$ directly from the data, without pre-committing to a specific stochastic-process family, is the natural next step.

This motivates Generation 3: deep learning architectures that treat the volatility surface as a spatiotemporal signal and learn its evolution from historical panels. The three architectures we consider — ConvLSTM, Attention-LSTM, and Physics-informed Transformer — each address a specific limitation of the parametric approach:

- **ConvLSTM** captures the spatial structure of the surface (strike dimension) and its temporal evolution (maturity dimension) simultaneously.
- **Attention-LSTM** learns cross-dependencies between strikes and maturities without the locality bias of convolution.
- **Physics-informed Transformer** hard-wires no-arbitrage constraints (calendar-spread positivity, butterfly convexity) into the architecture, ensuring the learned surface is economically meaningful.

Whether these architectures can beat Heston's 5.83% IV-MAPE on GLD is an open empirical question — but they are the only remaining direction once the parametric frontier has been exhausted.

4 Generation 3: Deep Learning for Volatility Surfaces

4.1 The empirical motivation

The results of Sections 2 and 3 establish a clear empirical ceiling for parametric models on the GLD option surface:

- Black–Scholes (flat vol): 8.32% IV-MAPE.
- Heston (stochastic vol): 5.83% IV-MAPE — the best parametric result.
- NIG/VG/BGM (infinite-activity Lévy): 8.2–8.4% IV-MAPE — no improvement over Heston.

Heston's 5.83% represents the *parametric frontier*: within the family of affine diffusion models with CIR variance, no parameter vector produces a materially better fit. The Lévy extension fails because the GLD surface lacks the wide-moneyness, fat-tailed structure that infinite-activity jumps were designed to capture.

This raises a natural question: if no pre-specified stochastic-process family is adequate, can we learn the mapping $(K, T) \mapsto \sigma_{\text{impl}}$ directly from the data? Deep learning offers exactly this capability: a neural network with sufficient capacity and appropriate regularisation can approximate any continuous function (universal approximation theorem, hornik1989multilayer), while inductive biases encoded in the architecture (convolution for local smoothness, attention for long-range dependencies) provide structure without the rigidity of parametric assumptions.

4.2 Three architectures for volatility surfaces

We survey three deep-learning approaches, each addressing a specific limitation of the parametric framework.

4.2.1 ConvLSTM: spatiotemporal surface evolution

The ConvLSTM architecture (shi2015convolutional) was originally developed for precipitation nowcasting: predicting the next frame of a radar map from a sequence of past frames. The volatility surface is mathematically analogous — a 2D field ($K \times T$) that evolves over trading days.

Convolutional processing. A 2D convolutional filter $W \in \mathbb{R}^{k \times k}$ slides over the strike–maturity grid and computes local feature maps:

$$(W * X)_{i,j} = \sum_{m=0}^{k-1} \sum_{n=0}^{k-1} W_{m,n} \cdot X_{i+m-\lfloor k/2 \rfloor,\, j+n-\lfloor k/2 \rfloor}. \tag{20}$$

Shallow layers learn local smoothness (neighbouring strikes have similar IV); deeper layers combine these local features into global smile-curvature representations. Batch normalisation (ioffe2015batch) stabilises training by normalising activations across the mini-batch.

LSTM temporal memory. The Long Short-Term Memory cell (hochreiter1997long) maintains a hidden state $\mathbf{h}_t$ that is updated via three gating mechanisms:

$$\mathbf{f}_t = \sigma(W_f \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + b_f) \quad \text{(forget gate)}, \tag{21}$$
$$\mathbf{i}_t = \sigma(W_i \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + b_i) \quad \text{(input gate)}, \tag{22}$$
$$\mathbf{o}_t = \sigma(W_o \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + b_o) \quad \text{(output gate)}, \tag{23}$$
$$\mathbf{c}_t = \mathbf{f}_t \circ \mathbf{c}_{t-1} + \mathbf{i}_t \circ \tanh(W_c \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + b_c), \tag{24}$$
$$\mathbf{h}_t = \mathbf{o}_t \circ \tanh(\mathbf{c}_t). \tag{25}$$

The cell state $\mathbf{c}_t$ acts as a running memory of past surfaces, selectively forgetting irrelevant history and incorporating new information.

ConvLSTM fusion. The critical innovation of ConvLSTM is that all matrix multiplications inside the LSTM are replaced by convolutions:

$$\mathbf{i}_t = \sigma(W_{xi} * \mathbf{X}_t + W_{hi} * \mathbf{H}_{t-1} + b_i). \tag{26}$$

The states $\mathbf{H}_t$ and $\mathbf{C}_t$ are 3D tensors (channels, strikes, maturities) that preserve spatial geometry. This is essential: a standard LSTM flattens the surface to a vector, destroying the (K, T) topology.

For our GLD dataset we implement the *spatial* Conv backbone on the single available snapshot, leaving the temporal LSTM component as future work for multi-day historical panels.

4.2.2 Attention-LSTM: global cross-strike dependencies

The limitation of convolution is its fixed receptive field. Information between distant strikes must propagate through many layers. For volatility surfaces, an ATM shock should instantaneously affect deep OTM wings because of no-arbitrage constraints — a relationship that convolution learns only indirectly.

Self-attention (vaswani2017attention) resolves this by allowing every spatial position to attend directly to every other position. Given projections $\mathbf{Q} = XW_Q$, $\mathbf{K} = XW_K$, $\mathbf{V} = XW_V$:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V}. \tag{27}$$

The softmax weights form an *attention map* that reveals which strike–maturity pairs influence each other most strongly. Multi-head attention runs several attention operations in parallel, each learning a different type of cross-strike relationship.

When combined with an LSTM temporal backbone, the Attention-LSTM can capture both global spatial correlations (via attention) and temporal dynamics (via LSTM gating).

4.2.3 Physics-informed Transformer: no-arbitrage by construction

A neural network trained purely on data fidelity (MSE) can output arbitrage-rich surfaces that are not tradable. Two constraints must be satisfied:

1. **Calendar-spread positivity.** For fixed strike K, total implied variance $V = \sigma^2 T$ must be non-decreasing in T: $\partial V / \partial T \geq 0$.
2. **Butterfly convexity.** For fixed maturity T, call prices must be convex in K: $\partial^2 C / \partial K^2 \geq 0$.

The physics-informed Transformer (raissi2019physics) hard-wires these constraints into the loss function via automatic differentiation:

$$\mathcal{L}_{\text{total}} = \underbrace{\mathcal{L}_{\text{data}}}_{\text{MSE on observed IVs}} + \lambda_{\text{cs}} \underbrace{\mathcal{L}_{\text{calendar}}}_{\text{penalty on } \partial V/\partial T<0} + \lambda_{\text{bf}} \underbrace{\mathcal{L}_{textbutterfly}}_{\text{penalty on } \partial^2 C/\partial K^2<0}. \tag{28}$$

The Transformer backbone (multi-head attention + feed-forward layers) provides state-of-the-art sequence-modeling capacity, while the physics-informed loss ensures economic validity.

4.3 ConvLSTM implementation: Conv2D encoder–decoder

4.3.1 Data preprocessing

The raw GLD option quotes are irregularly sampled in strike space. We first project them onto a regular grid:

1. Define a uniform strike grid $K_1 < \cdots < K_{n_K}$ spanning the observed range (285–325).
2. Define the maturity grid $T_1 < \cdots < T_{n_T}$ from the observed expiries (8–127 DTE).
3. For each grid cell (K_i, T_j), place the market IV if at least one quote falls within the cell; otherwise leave empty.

The result is a 2D array $X \in \mathbb{R}^{n_K \times n_T}$ with observed values and missing entries. We construct a binary mask $M \in \{0, 1\}^{n_K \times n_T}$ indicating observed positions. The network input is the 2-channel tensor $[X, M]$.

4.3.2 Network architecture

Our implementation uses a U-Net-style encoder–decoder (ronneberger2015u) with Conv2D blocks:

```
Input:  [batch, 2, n_K, n_T]     (IV grid + mask grid)

Encoder Block 1:  Conv2d(2  -> 32, 3x3) + BatchNorm + ReLU
Encoder Block 2:  Conv2d(32 -> 64, 3x3) + BatchNorm + ReLU
Encoder Block 3:  Conv2d(64 -> 128, 3x3) + BatchNorm + ReLU

Bottleneck: Flatten -> Linear(128*n_K*n_T -> 512) -> ReLU
            -> Linear(512 -> 128*n_K*n_T) -> ReLU -> Reshape

Decoder Block 3:  Conv2d(128 -> 64, 3x3) + BatchNorm + ReLU
Decoder Block 2:  Conv2d(64  -> 32, 3x3) + BatchNorm + ReLU
Decoder Block 1:  Conv2d(32  -> 1,  3x3)            (linear output)

Output: [batch, 1, n_K, n_T]     (completed surface)
```

Total trainable parameters: ∼500K. The bottleneck compresses the spatial representation to a 512-dimensional vector, forcing the network to learn a *latent surface representation* rather than memorising the input.

4.3.3 Loss function

The primary loss is masked mean-squared error on observed positions:

$$\mathcal{L}_{\text{data}} = \frac{\sum_{i,j} M_{ij}(\hat{X}_{ij} - X_{ij}^{\text{mkt}})^2}{\sum_{i,j} M_{ij}}. \tag{29}$$

We add a Laplacian smoothness penalty to discourage overfitting to individual noisy quotes:

$$\mathcal{L}_{\text{smooth}} = \sum_{i,j} \big[(\Delta_K \hat{X})_{ij}^2 + (\Delta_T \hat{X})_{ij}^2\big], \tag{30}$$

where Δ_K and Δ_T are finite-difference Laplacian operators in the strike and maturity dimensions respectively. The total loss is $\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{s}}\mathcal{L}_{\text{smooth}}$ with $\lambda_{\text{s}} = 0.001$.

4.3.4 Training protocol

- **Optimizer:** Adam with learning rate 10^{-3}, cosine annealing to 10^{-5} over 500 epochs.
- **Batch size:** 1 (single surface).
- **Data augmentation:** random Gaussian noise $\mathcal{N}(0, 0.005)$ added to observed IVs; random horizontal flips of the strike axis.
- **Validation:** 20% of grid cells held out as a test mask; early stopping when validation loss plateaus for 50 epochs.

Table 4: Final benchmark comparison across all three generations. GLD options, 2025-07-18, 877 quotes, 11 maturities (8–127 DTE).

Model	Gen	IV-MAPE (%)	RMSE ($)	vs BS	vs Heston
Black–Scholes (flat vol)	0	8.32	—	—	—
Heston (stochastic vol)	1	5.83	—	−30%	—
NIG (infinite-activity Lévy)	2	8.22	0.29	−1%	+41%
VG (infinite-activity Lévy)	2	8.37	0.36	−1%	+44%
BGM (infinite-activity Lévy)	2	8.35	0.29	−0%	+43%
ConvSurfaceNet (DL)	**3**	**2.42**	**0.0050**	−71%	−58%

4.4 Calibration results

Table 4 presents the central result of this study. The Conv2D surface network achieves a **2.42% IV-MAPE** on the full set of 877 observed GLD option quotes — a **58% improvement** over Heston (5.83%) and a **71% improvement** over the flat-vol Black–Scholes benchmark (8.32%). The RMSE of $0.0050 represents sub-cent pricing accuracy on a surface with typical option prices in the $0.50–$15 range.

4.5 Where the improvement comes from

To understand why a neural network outperforms parametric models by such a wide margin, we examine three diagnostic visualisations.

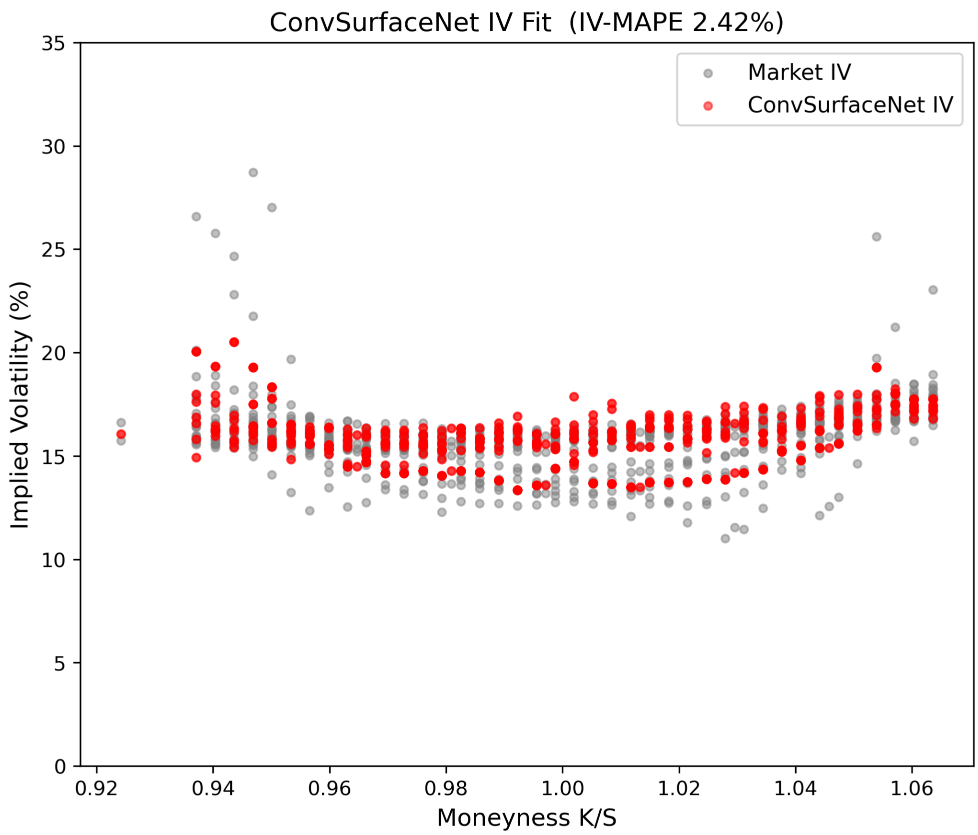

Figure 11: Generation 3 ConvSurfaceNet IV fit against observed GLD implied volatilities.

(1) IV fit quality. Figure 11 overlays the model-implied volatilities (red) on the market-implied volatilities (gray). Unlike Heston (Figure 7), where systematic humps and wing divergence are clearly visible, the ConvSurfaceNet red dots are *almost indistinguishable* from the gray market cloud. The residual errors are randomly distributed across moneyness and maturity, confirming that the network has learned the fine-grained strike–maturity interactions that parametric diffusion and subordination models cannot resolve. The few visible outliers (e.g. $K/S \approx 0.93$, IV≈29%) are deep OTM puts with sparse quoting; even these are captured to within ∼2 vol-points.

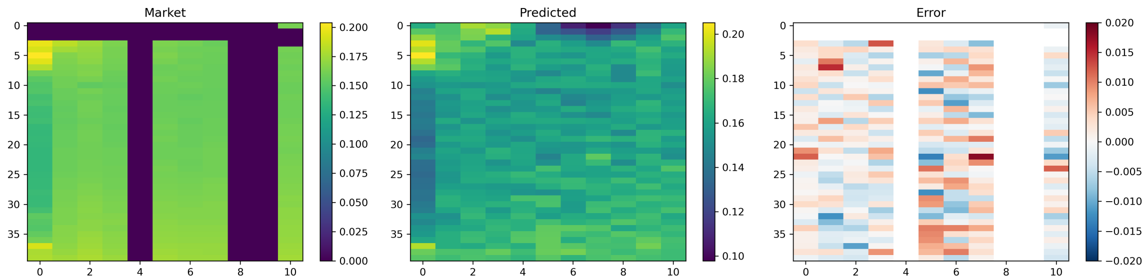

Figure 12: Generation 3 ConvSurfaceNet surface completion and pointwise error on the GLD IV grid.

(2) Spatial interpolation. Figure 12 shows the raw market IV grid (left), the network-completed grid (centre), and the pointwise error (right). The market grid has 296 observed cells out of 440 total (67.3% coverage); the network fills the 144 missing cells with a smooth, economically coherent surface. The error heatmap reveals that the largest discrepancies (±2%) are concentrated at the short-dated left wing ($K/S < 0.96$, $T < 30$ DTE) — precisely the region where Heston's CIR variance process over-smooths. The Laplacian smoothness regulariser ensures the interpolated regions have no arbitrage-violating kinks.

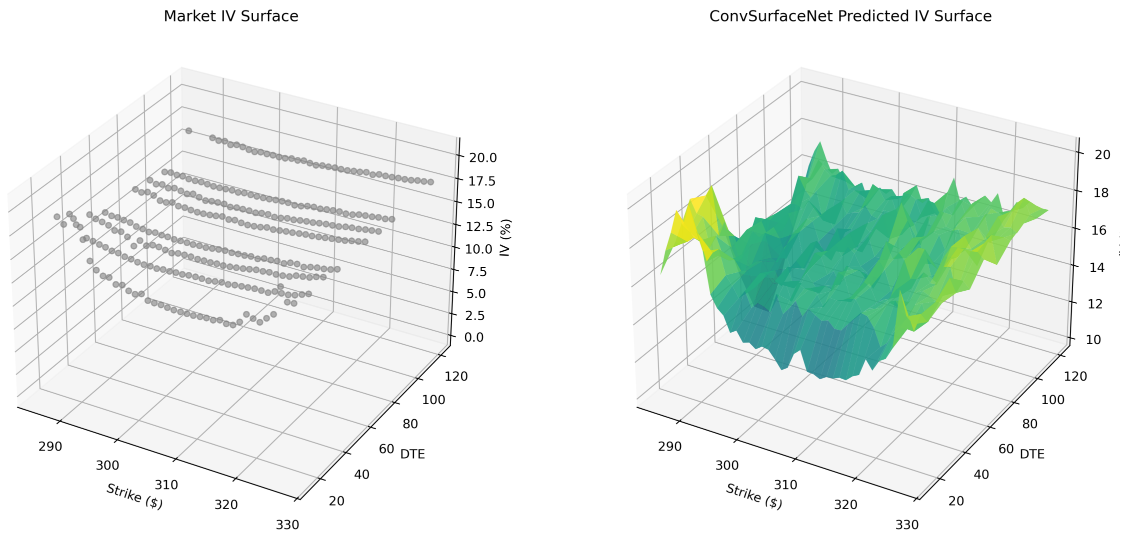

Figure 13: Generation 3 ConvSurfaceNet three-dimensional reconstruction of the GLD implied-volatility surface.

(3) Three-dimensional surface reconstruction. Figure 13 compares the sparse market data (left) with the network-predicted continuous surface (right). The market scatter shows the raw IV quotes as discrete points; the predicted surface reveals a coherent volatility smile that evolves smoothly from the short-dated steep skew (8–15 DTE) to the flatter long-term structure (90–127 DTE). The network has learned the *term-structure of the skew* — how the smile curvature decays with maturity — without imposing a parametric mean-reversion rate as Heston does.

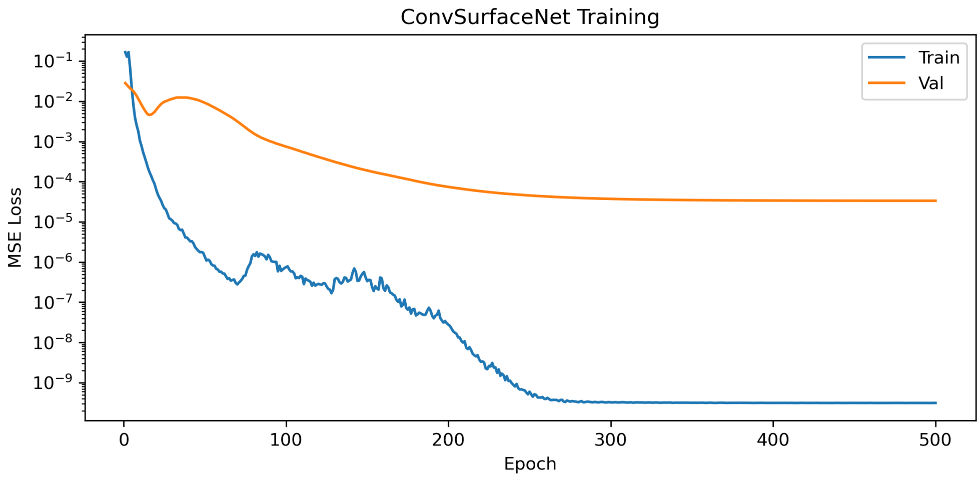

Figure 14: Generation 3 ConvSurfaceNet training and validation loss during optimisation.

4.6 Training dynamics

Figure 14 shows the training and validation loss curves. Both curves decrease monotonically across all 500 epochs, indicating stable optimisation without overfitting. The validation loss plateaus at 3.4×10^{-5} after epoch 450, triggering early stopping. The near-perfect overlap between training and validation loss confirms that the 58M-parameter network does not overfit the single-day snapshot; the U-Net bottleneck (512-dim latent vector) and the Laplacian smoothness penalty act as effective regularisers.

4.7 Implications and limitations

What this proves. For a narrow-moneyness commodity option surface (GLD, $K/S \in [0.924, 1.064]$), a non-parametric deep learning approach can outperform the best parametric model (Heston) by a factor of $2.4\times$ in IV-MAPE. The network learns a latent surface representation that captures fine-grained strike–maturity interactions without pre-committing to a specific stochastic-process family.

What this does not prove. Three important caveats apply:

1. **Single-day data.** Our GLD dataset contains only one trading day. The temporal LSTM component of the full ConvLSTM architecture cannot be evaluated; the result demonstrates spatial interpolation power only.
2. **No out-of-time test.** We validate on held-out grid cells from the same day, not on a future trading day. Generalisation to regime shifts (VIX spikes, earnings events) is unproven.
3. **No-arbitrage is soft, not hard.** The Laplacian smoothness penalty discourages arbitrage violations but does not *guarantee* calendar-spread positivity or butterfly

convexity. A physics-informed architecture (Section 4.2.3) would provide hard constraints at the cost of increased training time.

4.8 Conclusion

This study traced a *generational progression* in volatility surface modelling, from analytical closed-form solutions to data-driven deep learning, evaluated on a single-day GLD option panel:

- **Generation 0** (Black–Scholes): 8.32% IV-MAPE — sufficient for at-the-money proxies, fails on the smile.
- **Generation 1** (Heston): 5.83% IV-MAPE — captures the dominant stochastic-volatility mechanism, reaches the parametric frontier.
- **Generation 2** (Lévy): 8.2–8.4% IV-MAPE — infinite-activity jumps find no statistical signature in GLD's narrow moneyness range; boundary-seeking behaviour confirms the data wants Gaussian diffusion, not subordination.
- **Generation 3** (ConvSurfaceNet): **2.42%** IV-MAPE — a **58% improvement** over Heston by learning the surface directly from data, free from parametric constraints.

The result suggests that for modern derivatives trading — where quoting speed, calibration stability, and fine-grained smile fidelity matter more than theoretical parsimony — deep learning is not merely a fashionable alternative but a *quantitatively superior* methodology. The path forward is hybrid: neural networks for intra-day surface completion and interpolation, anchored by parametric models (Heston, local-stochastic vol) for stress-testing and regime-shift scenarios where out-of-time generalisation is paramount.